25901894 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCMA200 - AI

TECHNIQUES FOR DATA ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Differentiate between nominal and ordinal data. [U] / [CO1]
2. List two applications of AI in manufacturing. [K] / [CO1]
3. Write a Python example code using the break statement in a loop. [U] / [CO2]
4. Give two differences between Numpy array and Python list. [U] / [CO2]
5. Define Pandas Index. What is the default index in a Pandas DataFrame? [K] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Write briefly about some popular data analytics tools with their key uses. [K] / [CO1]
7. Explain the growth of AI from 2010 to present. [K] / [CO1]

8. Explain the importance of data structures in Python with examples. [K] / [CO2]

9. Discuss relational operators in Python with examples. [U] / [CO2]

10. Describe different methods to detect outliers in a dataset with examples. [U] / [CO3]

11. Differentiate the use of line plot and bar plot. [U] / [CO3]

12. Differentiate between descriptive statistics functions mean(), median(), and mode() in Pandas. [U] / [CO3]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Explain selective statements in Python with examples. [U] / [CO2]

14. What is data visualization? Explain the importance of visualization in analytics and discuss different types of charts/graphs with examples. [U] / [CO3]

[15x1 = 15]